

Factor Models of Interest Rate Curves

Homework Assignment 1 — MTH 9877, Group E

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March 2026

Outline

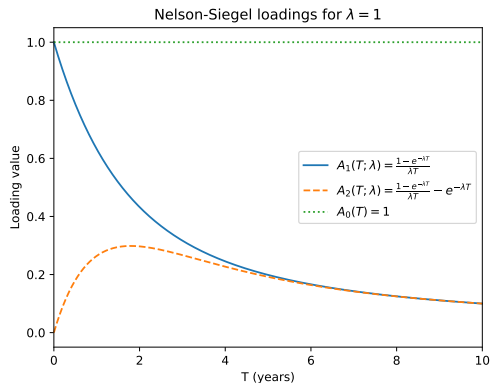
- 1 Part A: Nelson–Siegel Model
- 2 Part B: Data
- 3 Part C: Cross-Sectional Calibration
- 4 Part D1: Dynamic NS via Kalman Filter
- 5 Part E: Applications
- 6 Conclusion

The Nelson–Siegel forward rate is $y(s) = \beta_0 + \beta_1 e^{-\lambda s} + \beta_2 \lambda s e^{-\lambda s}$. The zero-coupon yield is

$$Y(T) = \frac{1}{T} \int_0^T y(s) ds = \beta_0 + \beta_1 \underbrace{\frac{1 - e^{-\lambda T}}{\lambda T}}_{A_1} + \beta_2 \underbrace{\left(\frac{1 - e^{-\lambda T}}{\lambda T} - e^{-\lambda T} \right)}_{A_2}$$

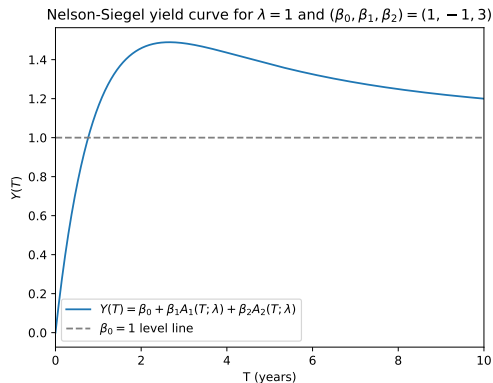
- β_0 long-run level ($Y(\infty) = \beta_0$)
- β_1 short-end adjustment ($Y(0^+) = \beta_0 + \beta_1$)
- β_2 mid-maturity curvature (hump or dip)
- λ timescale (hump location $\approx 1/\lambda$)

NS Loadings and Example Curve



A_1 : monotone decay from 1 to 0.

A_2 : hump-shaped, peaks near $T \approx 1/\lambda$.



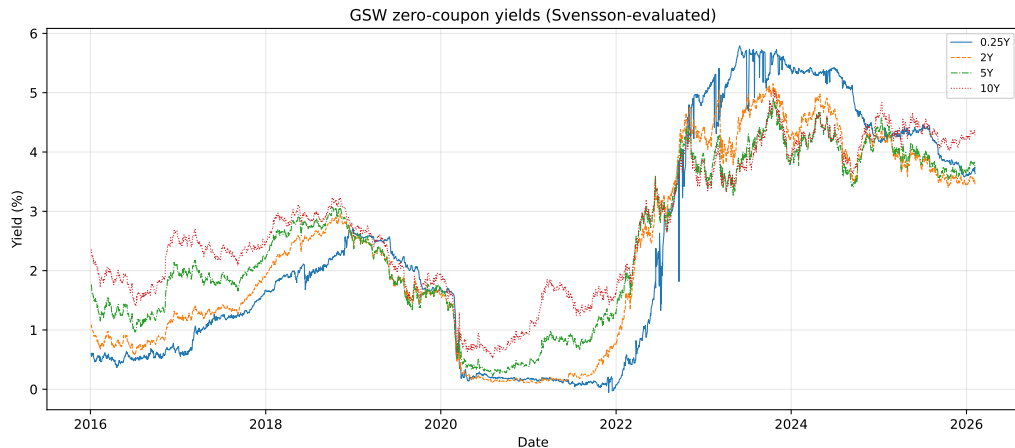
Example: $(\beta_0, \beta_1, \beta_2) = (1, -1, 3)$, $\lambda = 1$.

Upward slope from $\beta_1 < 0$; hump from $\beta_2 > 0$.

Part B: Data

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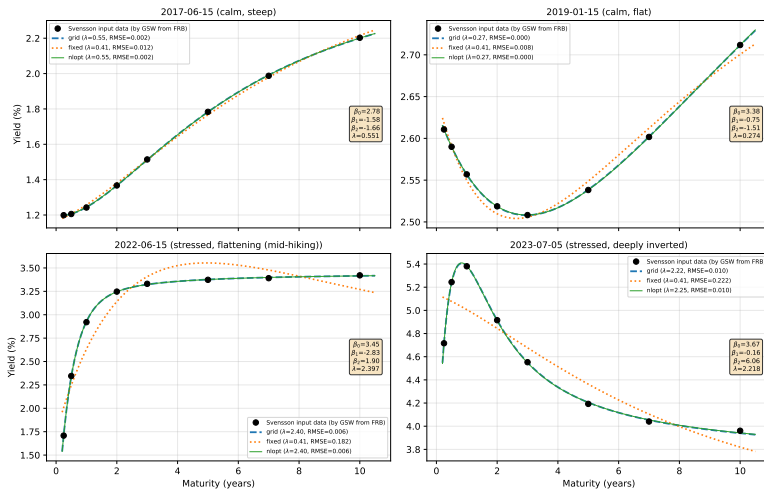
GSW Daily Yield Panel (2016–2026)



2,524 daily observations across 8 maturities, spanning hiking, COVID near-zero, inversion, and easing regimes.

NS Fit on Selected Dates

Observed vs. Nelson-Siegel fit (three lambda approaches)

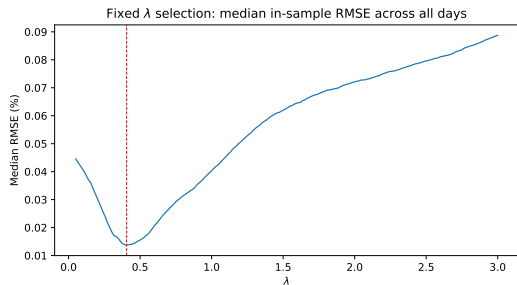


The inputs and the fits discussed later.

Part C: Cross-Sectional Calibration

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Fit Quality: Three λ Methods

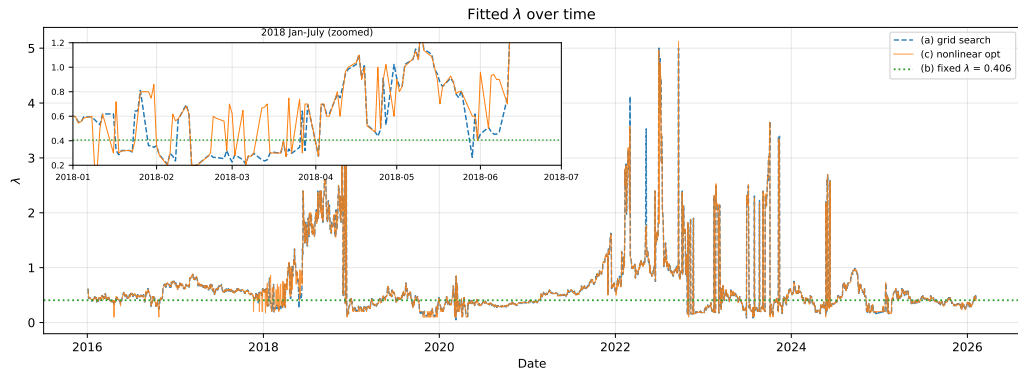


Method	Med	Mean	Max
(a) Grid search	0.0016	0.0053	0.0615
(b) Fixed λ^*	0.0137	0.0261	0.5009
(c) Nonlinear opt	0.0015	0.0053	0.0615

RMSE (%)

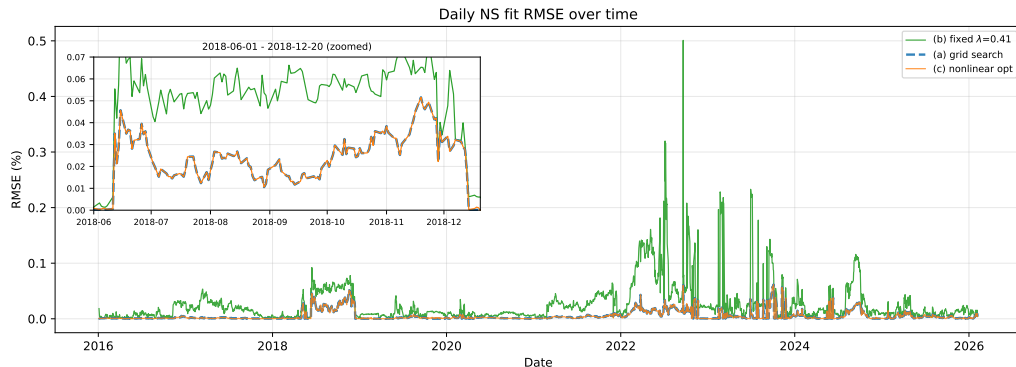
Grid search and nonlinear opt are near-identical; fixed λ adds no cost in calm regimes but fails badly under stress.

Time Series of λ



The daily λ path tracks the Fed's policy regime — shifting markedly higher during the 2022–23 inversion.

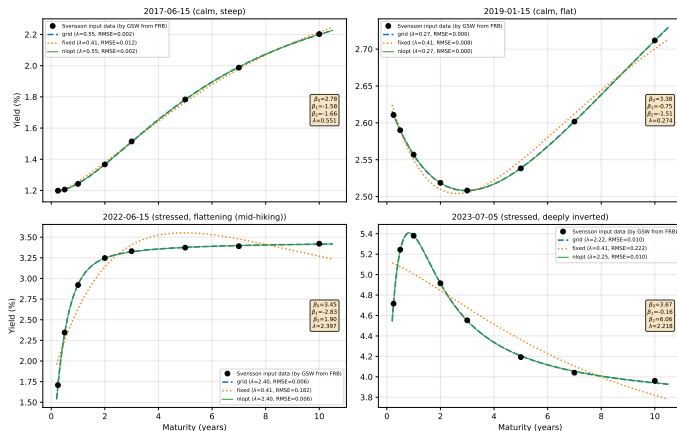
Daily RMSE: Three Methods



Fixed λ RMSE spikes to $\sim 0.5\%$ during 2022–23 — roughly two orders of magnitude above the adaptive methods.

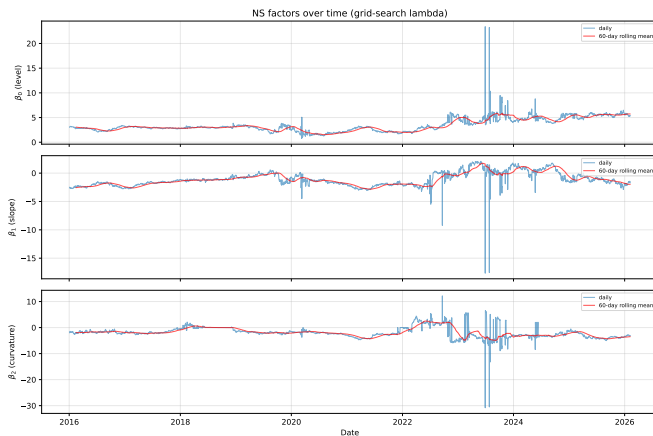
Representative Curve Fits

Observed vs. Nelson-Siegel fit (three lambda approaches)



Fixed $\lambda = 0.41$ is $30\times$ worse on 2022-06-15 and $20\times$ worse on 2023-07-05 compared to adaptive methods.

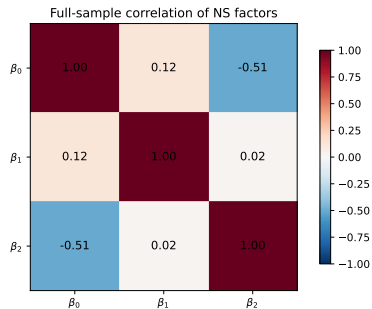
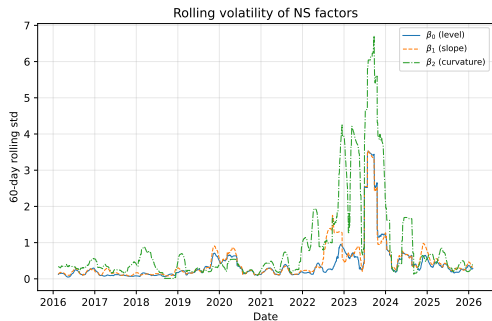
NS Factor Time Series



	Min	Max
β_0	0.74	23.41
β_1	-17.64	2.09
β_2	-30.84	12.18

Extreme spikes in 2022–23 are numerical artefacts from multiple near-equivalent OLS solutions — addressed by the Kalman filter in Part D.

Factor Volatility and Correlation



Volatility spikes sharply in 2022–23 then reverts; β_0 and β_2 are moderately negatively correlated (-0.51).

Parameter Instability: $\beta_0 > 20$ Outliers

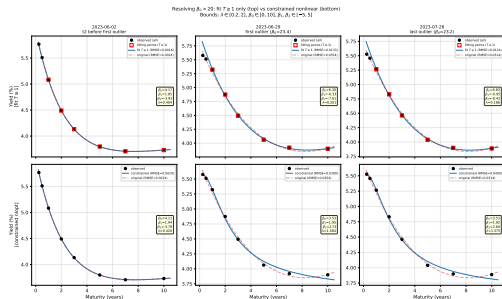
Two remedies tested:

- 1 Restrict fitting to $T \geq 1$ Y (drop 0.25 and 0.5 Y)
- 2 Constrained nonlinear: $\lambda \in [0.2, 2]$, $\beta_0 \in [0, 10]$, $\beta_1, \beta_2 \in [-5, 5]$

Both eliminate $\beta_0 > 20$ but introduce trade-offs:

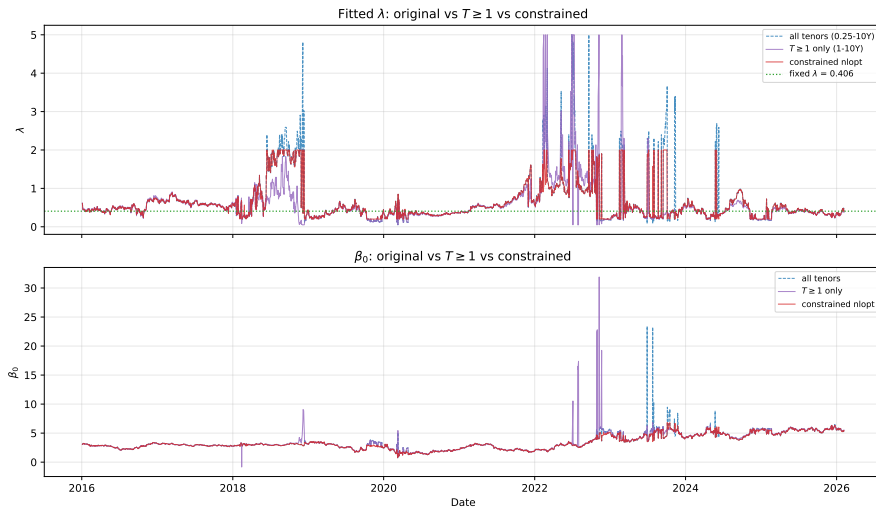
- $T \geq 1$: loses sub-year maturities
- Constrained: slightly higher RMSE on outlier dates

Neither addresses the root cause (identification); the Kalman filter in Part D provides a principled solution.



NS fit on the two outlier dates after applying the remedies.

λ and β_0 : Before and After Remedies

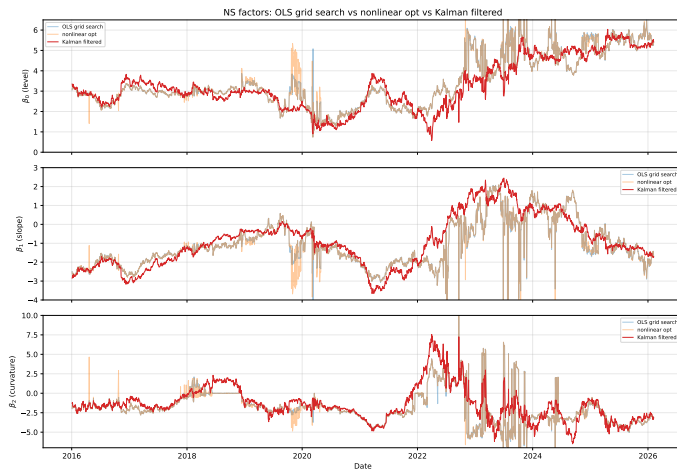


Time series of λ (top) and β_0 (bottom) from grid search: original vs $T \geq 1$ vs constrained.

Part D1: Dynamic NS via Kalman Filter

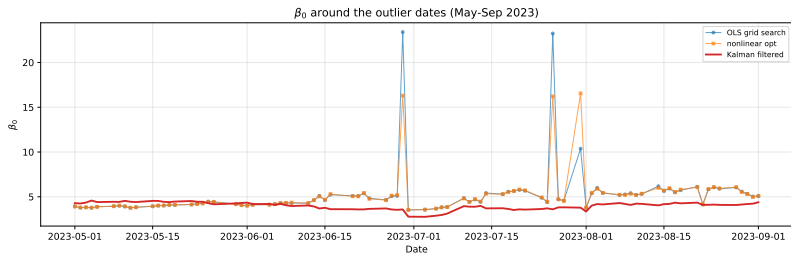
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Filtered vs. OLS vs. Nonlinear Factors



The VAR(1) prior rejects pathological solutions automatically — no ad-hoc constraints required.

β_0 Around the Outlier Period (May–Sep 2023)

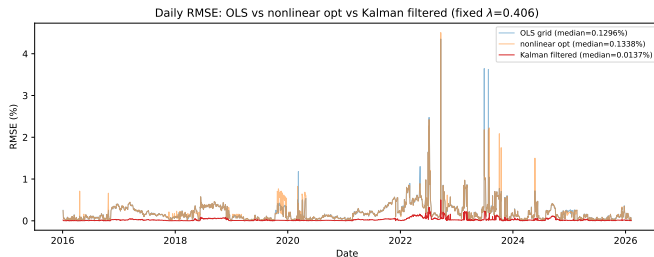


Ranges for β_0 across the three methods for the whole sample:

Method	Min	Max
OLS grid	0.74	23.40
Nonlinear	0.74	16.55
Kalman	0.57	6.04

The Kalman filter stays on the economically sensible solution branch throughout.

In-Sample RMSE: OLS vs. Kalman

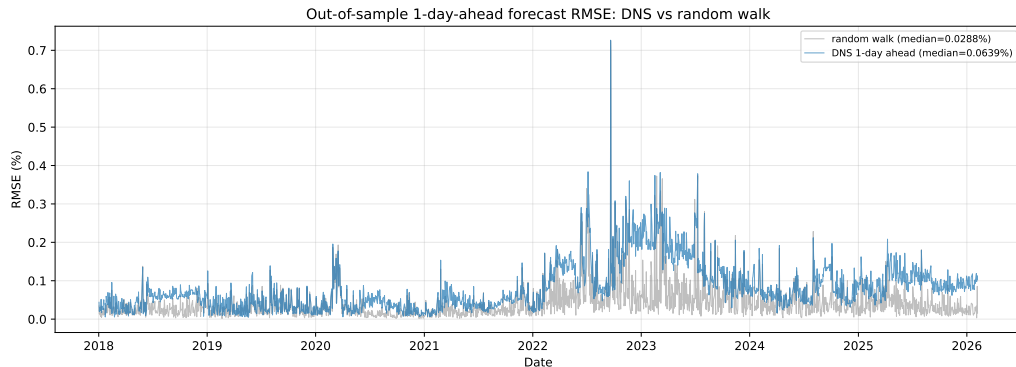


	Med	Mean	Max
OLS	0.130	0.190	4.35
Nlopt	0.134	0.195	4.51
Kalman	0.014	0.026	0.50

RMSE (%)

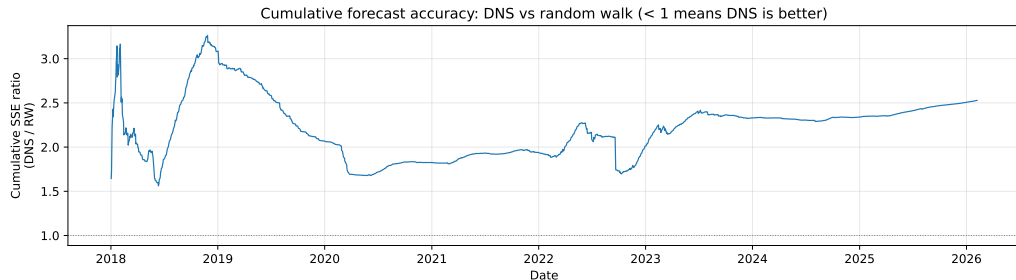
Kalman achieves $10\times$ lower median RMSE and $8.7\times$ lower worst-case RMSE by staying on the better-fitting solution.

Out-of-Sample Forecast RMSE: DNS vs. Random Walk



DNS median RMSE (0.064%) is more than twice the random walk (0.029%) at the 1-day horizon.

Cumulative Forecast Accuracy: DNS vs. Random Walk

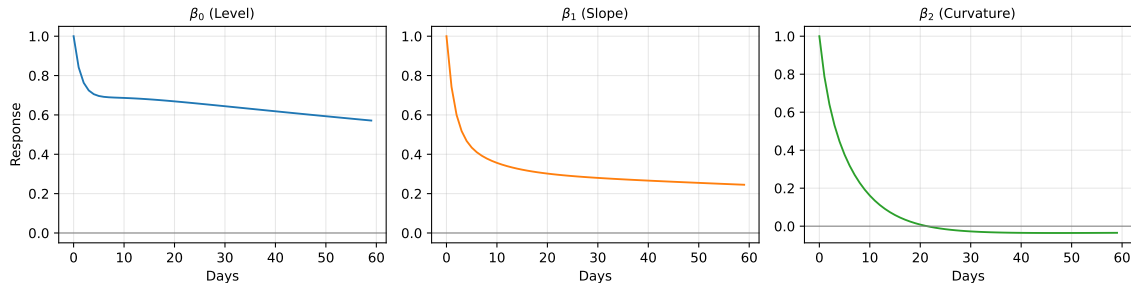


	Med RMSE	SSE ratio
RW	0.0288%	1.00
DNS	0.0639%	2.53

Dominant eigenvalue 0.9958 — the VAR barely differs from a random walk at 1-day horizon, and its mean-reversion prior is counterproductive during the 2022–23 trending regime.

VAR(1) Impulse Response Functions

VAR(1) own-factor impulse responses (diagonal)



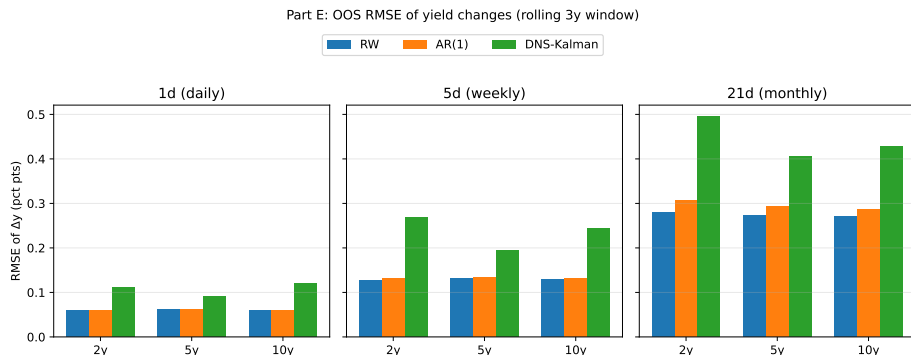
The three NS factors exhibit a clear persistence hierarchy — level shocks are nearly permanent (half-life 166 days), slope decays over weeks, and curvature reverts to zero within days.

Part E: Applications

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Part E: Forecasting yield changes and trading

Design. Out-of-sample forecasts of *changes* in 2y, 5y, and 10y zero-coupon yields at 1-, 5-, and 21-day horizons; rolling three-year estimation window



Result. Random walk and AR(1) achieve lower RMSE than DNS–Kalman uniformly; we find no empirical basis for DNS-driven directional trading versus these benchmarks.

- ① **Grid search and nonlinear optimisation are equivalent** (median RMSE 0.0016%); joint optimisation adds no value over a fine grid.
- ② **Fixed λ fails under stress.** All methods are comparable in calm regimes, but during the 2022–2023 inversion the fixed- λ RMSE spikes up to $30\times$ above the adaptive methods.
- ③ **The Kalman filter improves factor quality.** VAR(1) dynamics eliminate OLS outliers and reduce median in-sample RMSE $10\times$ (0.0137% vs. 0.1296%).
- ④ **Filtered factors do not improve short-horizon forecasting.** The DNS Kalman filter is outperformed by the random walk at the short-term horizon; filtered factors are better suited to curve interpolation, risk factor extraction, and derivatives pricing.

Thank You